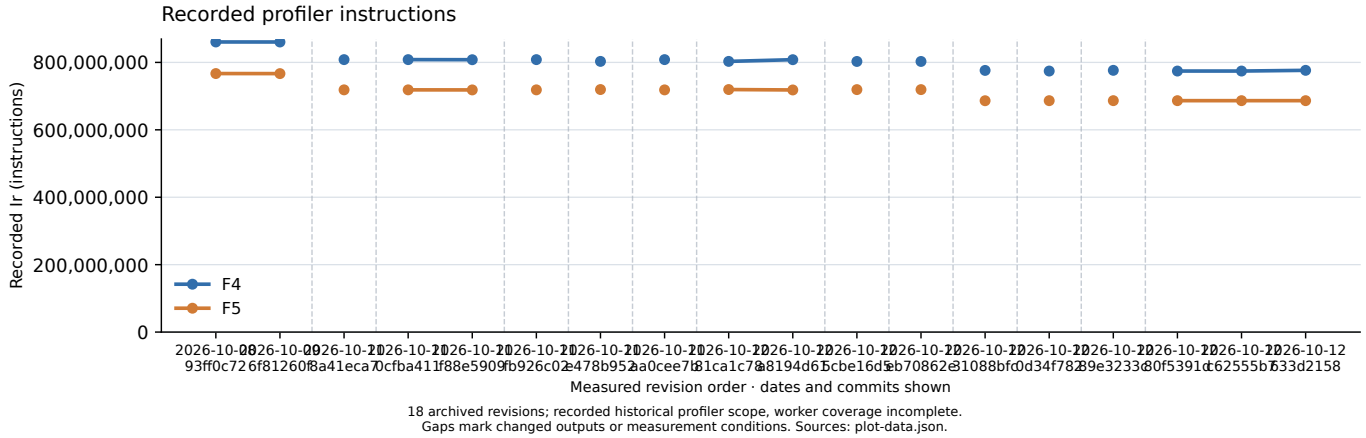
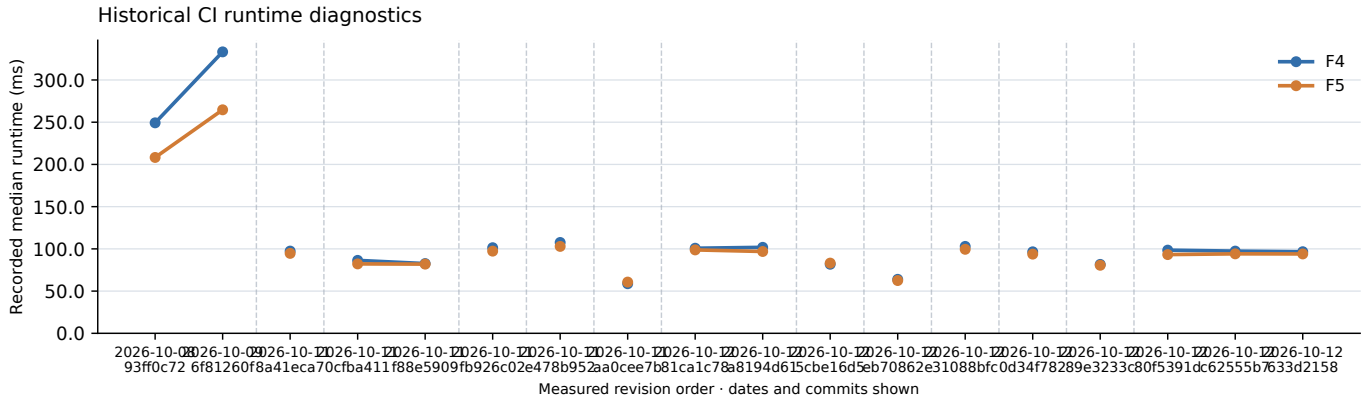
# Boolean matrix step: F4 and F5

One shared K_1 / 2^23 m=2 system, 22 Boolean variables, maximum degree 4. Matrix construction, reduction and readback; F5 includes its criterion.




18 archived revisions; recorded historical profiler scope, worker coverage incomplete.
Gaps mark changed outputs or measurement conditions. Sources: plot-data.json.




18 archived revisions; recorded medians, no stored spread or isolated timing eligibility.
CI runtime diagnostics; sample counts and condition changes retained in plot-data.json.

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-08 / 93ff0c72 | 860,633,889 | 766,940,469 | 249.241 | 208.250 |
| 2026-10-09 / 6f81260f | 860,555,567 | 766,776,087 | 333.279 | 264.687 |
| 2026-10-11 / 8a41eca7 | 808,263,705 | 718,576,235 | 97.380 | 94.658 |
| 2026-10-11 / 0cfba411 | 808,263,705 | 718,578,523 | 86.390 | 82.214 |
| 2026-10-11 / f88e5909 | 807,989,935 | 718,391,055 | 82.597 | 81.948 |
| 2026-10-11 / fb926c02 | 808,263,705 | 718,576,189 | 101.379 | 97.323 |
| 2026-10-11 / e478b952 | 803,017,657 | 719,516,063 | 107.585 | 102.927 |
| 2026-10-11 / aa0cee7b | 808,281,869 | 718,492,761 | 58.723 | 60.673 |
| 2026-10-12 / 81ca1c78 | 803,015,877 | 719,509,003 | 100.762 | 98.775 |
| 2026-10-12 / a8194d61 | 808,072,539 | 718,282,030 | 101.827 | 96.904 |
| 2026-10-12 / 5cbe16d5 | 802,824,065 | 719,316,911 | 81.784 | 83.144 |
| 2026-10-12 / eb70862e | 802,824,065 | 719,316,876 | 63.944 | 62.505 |
| 2026-10-12 / 31088bfc | 776,242,363 | 686,315,894 | 102.929 | 99.489 |
| 2026-10-12 / 0d34f782 | 774,379,185 | 686,521,131 | 96.521 | 93.790 |
| 2026-10-12 / 89e3233c | 776,490,367 | 686,543,371 | 81.628 | 80.581 |
| 2026-10-12 / 80f5391d | 774,379,185 | 686,519,896 | 98.501 | 93.239 |
| 2026-10-12 / c62555b7 | 774,379,185 | 686,519,896 | 97.360 | 94.117 |
| 2026-10-12 / 633d2158 | 776,467,358 | 686,512,000 | 96.637 | 94.008 |

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

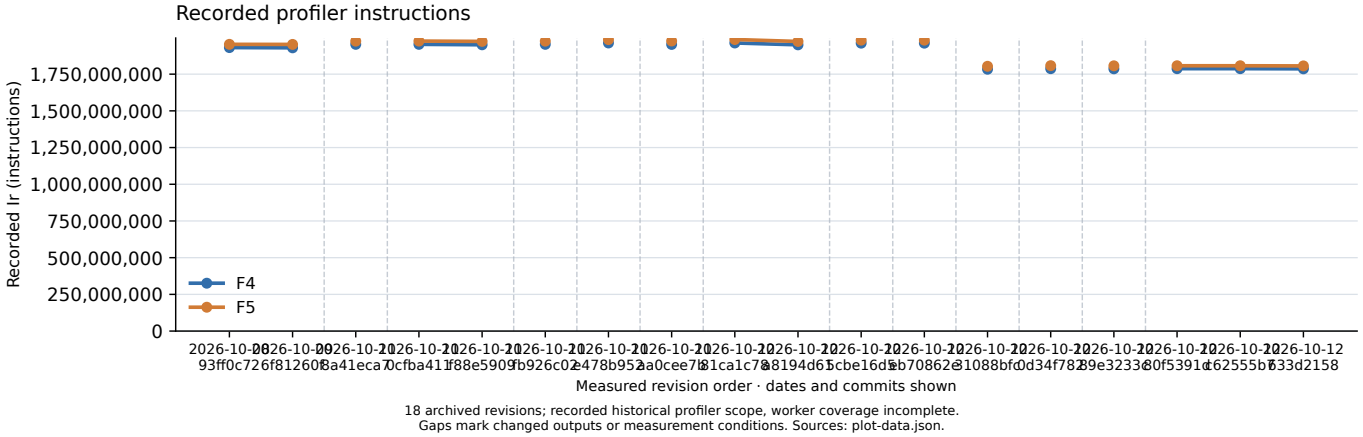
Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

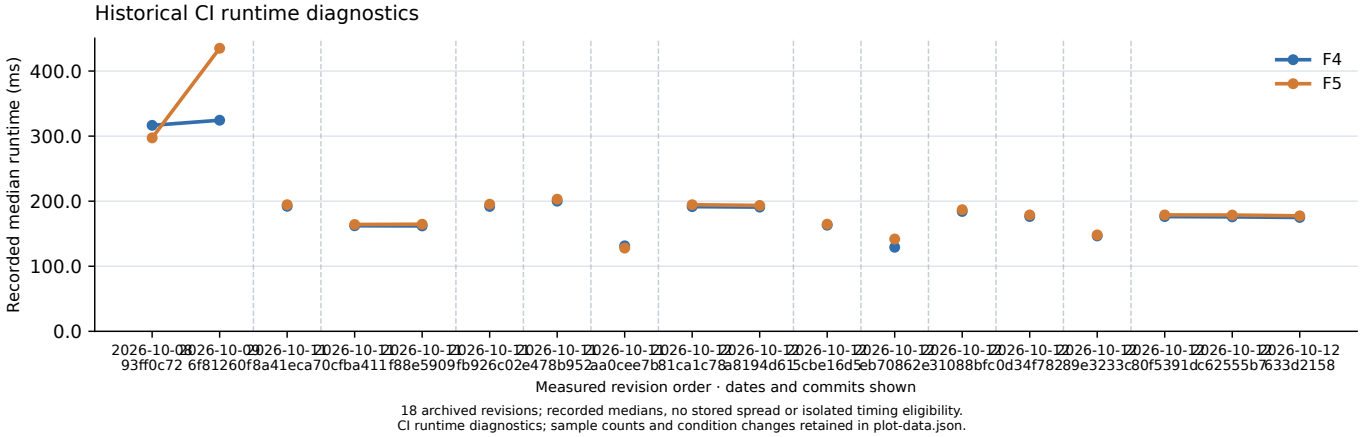
X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

Evidence: 18 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

# Boolean solver calls: F4 and F5

Shared K_1 / 2^17 m=2 workload, 16 Boolean variables, four planted and four random targets, maximum degree 3. The solve_boolean_system entry point, including root readback.




| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-08 / 93ff0c72 | 1,930,989,200 | 1,953,108,992 | 316.570 | 297.297 |
| 2026-10-09 / 6f81260f | 1,929,374,570 | 1,952,598,745 | 324.403 | 435.004 |
| 2026-10-11 / 8a41eca7 | 1,953,021,023 | 1,974,789,730 | 192.203 | 194.561 |
| 2026-10-11 / 0cfba411 | 1,953,020,766 | 1,975,267,516 | 162.266 | 164.161 |
| 2026-10-11 / f88e5909 | 1,949,981,977 | 1,973,219,339 | 161.982 | 164.584 |
| 2026-10-11 / fb926c02 | 1,953,020,766 | 1,974,789,781 | 191.988 | 195.374 |
| 2026-10-11 / e478b952 | 1,962,376,386 | 1,985,422,672 | 200.058 | 202.996 |
| 2026-10-11 / aa0cee7b | 1,952,402,641 | 1,975,976,609 | 131.376 | 127.933 |
| 2026-10-12 / 81ca1c78 | 1,962,376,386 | 1,985,933,163 | 191.630 | 194.567 |
| 2026-10-12 / a8194d61 | 1,949,694,281 | 1,972,772,104 | 190.899 | 193.455 |
| 2026-10-12 / 5cbe16d5 | 1,961,124,313 | 1,981,687,200 | 163.135 | 164.619 |
| 2026-10-12 / eb70862e | 1,961,124,313 | 1,981,686,883 | 129.169 | 141.847 |
| 2026-10-12 / 31088bfc | 1,783,643,232 | 1,803,568,510 | 184.204 | 186.912 |
| 2026-10-12 / 0d34f782 | 1,787,314,371 | 1,807,797,314 | 176.437 | 178.824 |
| 2026-10-12 / 89e3233c | 1,786,351,448 | 1,806,842,750 | 146.640 | 148.159 |
| 2026-10-12 / 80f5391d | 1,787,314,371 | 1,807,244,283 | 176.353 | 178.878 |
| 2026-10-12 / c62555b7 | 1,787,314,371 | 1,807,244,283 | 175.899 | 178.689 |
| 2026-10-12 / 633d2158 | 1,786,234,678 | 1,806,168,901 | 174.990 | 177.494 |

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

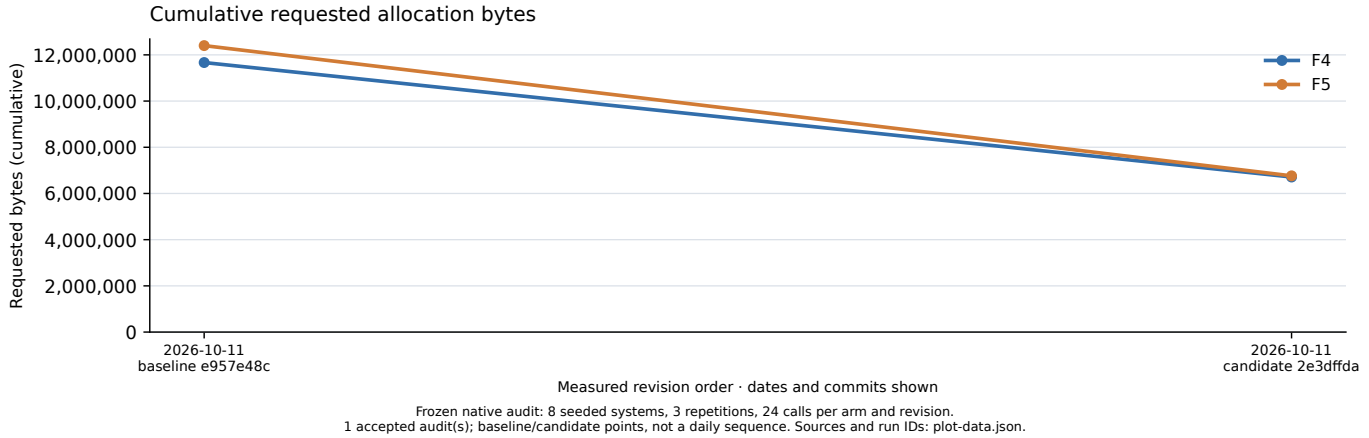
Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

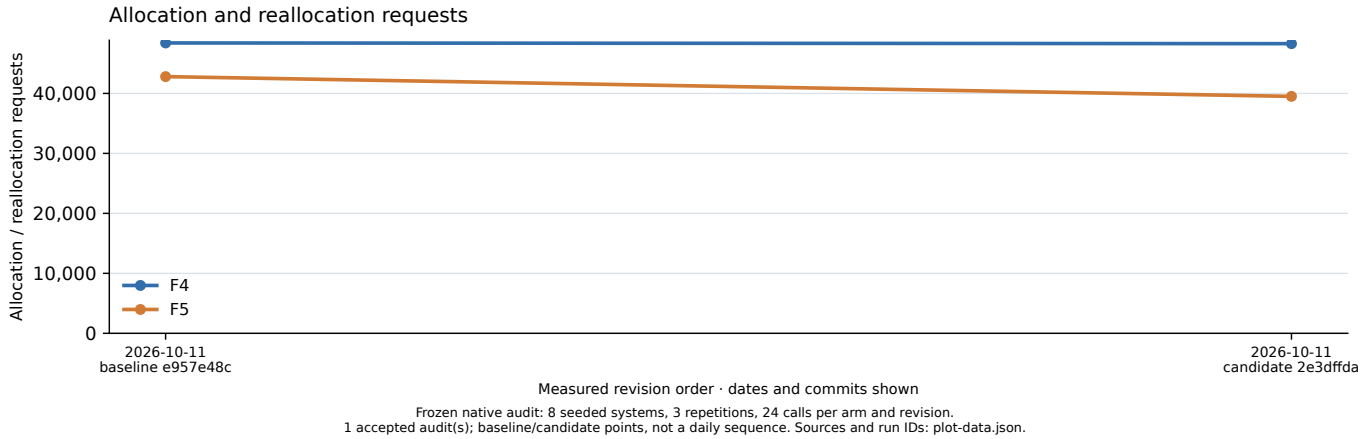
X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

Evidence: 18 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

## Complete Boolean calls: allocation history

Matching frozen Boolean allocation suite: 8 seeded systems, 3 repetitions, 24 calls per arm and revision. Polynomial dimensions, degrees and seeds are retained in the source audit protocol.




| UTC date / revision | F4 bytes | F5 bytes | F4 requests | F5 requests |
|---|---|---|---|---|
| 2026-10-11 / e957e48c | 11,667,696 | 12,400,920 | 48,414 | 42,801 |
| 2026-10-11 / 2e3dffda | 6,712,608 | 6,762,696 | 48,294 | 39,507 |

Native allocation audit. Original baseline/candidate rows, accepted comparison, exact outputs and accounting, shared inputs and all totals revalidated by the Rust plot-data tool.

One baseline and one candidate point per accepted audit, not a daily sequence. A/A and replication pairs are controls, not additional algorithm revisions.

Bytes are cumulative requested traffic, not peak memory. No timing is measured by this allocation series.

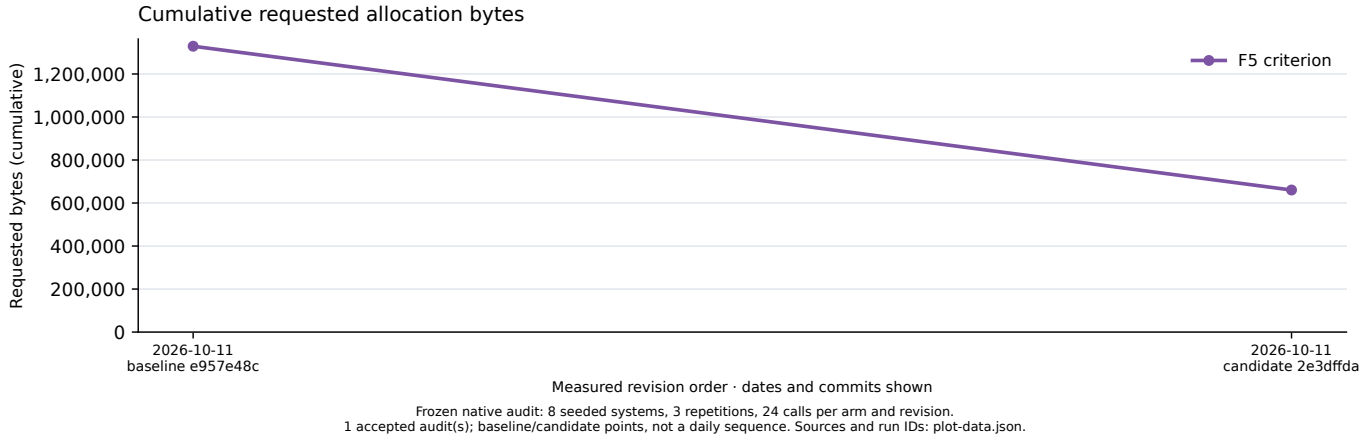
Criterion construction is a separate part of F5 and is never compared as a complete solver.

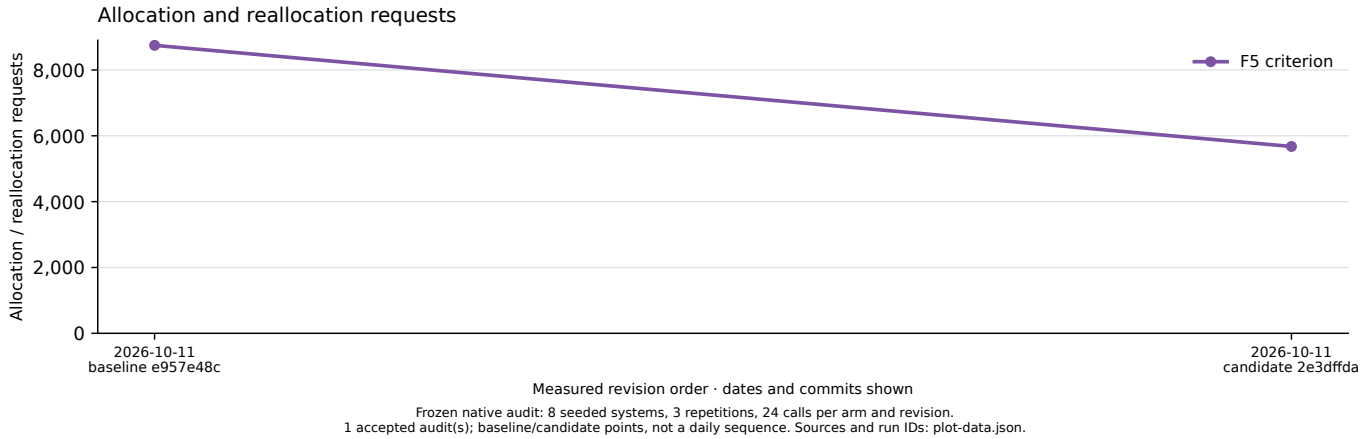
Audit revisions come from their frozen records. They are not inserted as measurements of historical CI workloads; the workloads and metrics remain separate.

Evidence: allocation_sources in plot-data.json; frozen native comparison and baseline/candidate JSONL records in research/f4_f5_row_build_20261011/records/. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

# F5 criterion: allocation history

Matching frozen Boolean allocation suite: 8 seeded systems, 3 repetitions, 24 calls per arm and revision. Polynomial dimensions, degrees and seeds are retained in the source audit protocol.




| UTC date / revision | F5 criterion bytes | F5 criterion requests |
|---|---|---|
| 2026-10-11 / e957e48c | 1,329,024 | 8,748 |
| 2026-10-11 / 2e3dffda | 660,480 | 5,676 |

Native allocation audit. Original baseline/candidate rows, accepted comparison, exact outputs and accounting, shared inputs and all totals revalidated by the Rust plot-data tool.

One baseline and one candidate point per accepted audit, not a daily sequence. A/A and replication pairs are controls, not additional algorithm revisions.

Bytes are cumulative requested traffic, not peak memory. No timing is measured by this allocation series.

Criterion construction is a separate part of F5 and is never compared as a complete solver.

Audit revisions come from their frozen records. They are not inserted as measurements of historical CI workloads; the workloads and metrics remain separate.

Evidence: allocation_sources in plot-data.json; frozen native comparison and baseline/candidate JSONL records in research/f4_f5_row_build_20261011/records/. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.